LedgerProof — return briefing

Prepared October 24, 2026, America/Los_Angeles. This is the index for the autonomous work package.

LedgerProof checks whether the invoices behind a cash forecast are trustworthy. The demo agency looks comfortable at $16,000 after two payrolls; an approved credit reduces that to $8,000, and a collection delay reveals a $4,000 shortfall. The useful product is the chain from source evidence to a reviewed correction to an understandable cash decision.

Current state: working local rehearsal; mandatory hosted CREAO execution and Convex deployment are still blocked by account setup. Code and diagrams do not establish live integration or sponsor eligibility.

Read or open these first

Artifact	What you get
Interactive briefing	Self-contained offline guide, diagrams, service search, and roadmap tabs; no account or server required
Printable briefing	Portable report for reviewing or sharing after checking its contents
Architecture atlas	Editable Mermaid plus standalone SVG/PNG architecture, lifecycle, ownership, and roadmap diagrams
How it works	Full user flow, cash calculations, code paths, design choices, failure behavior, and service responsibilities
Service register	Exactly what is built, configured, tested, blocked, and deliberately excluded for each service
Future development	Sequenced technical/product roadmap, new expansion paths, metrics, commercial hypotheses, and stop conditions
Event strategy	Updated guide interpretation, newly discovered tracks, fit ranking, evidence gaps, and schedule
Your action list	Prioritized actions that need your accounts, eligibility information, or publishing/submission decisions
Evaluation	Actual executed checks; offline evaluation separated from live provider verification

Local use

```
npm ci
npm run fixtures
npm run dev
```

Open http://localhost:3000 in LedgerProof. Start demo → upload the generated credit → open Investigation activity → explicitly select local rehearsal → run → inspect evidence → approve credit → move collection to November 9. Select weeks 2 and 3 to trace exact cash movements. Reset only your own workspace. See DEMO.md.

For the briefing alone, open docs/briefing.html directly. Diagrams are under docs/diagrams. Rebuild documentation assets with `npm run docs:build`; export PDF/PNG with `npm run docs:export` after installing Playwright Chromium.

Autonomous changes made in this pass

Added the architecture atlas, service inventory, operating explanation, development strategy, event/source register, stage log, submission checklist, and an offline navigable/printable briefing. Reviewed the user-supplied event guide and retrieved the linked Coinbase and Figma challenge content through Google Drive.

Hardened CREAO configuration: one explicitly selected API family, no automatic cross-family fallback, family-specific request/response validation, structured text handling, bounded payloads, wrong-run rejection, and malformed outputs escalated to review. Both adapters are contract-tested with mock HTTP responses, **not live verified**.

Corrected a finance edge case found during architecture review: newly posted receipts now enter the forecast in their actual date bucket, rather than all being added to opening cash. Source amounts with fractional dollars now retain cents in the display. The exact judge cash story remains unchanged.

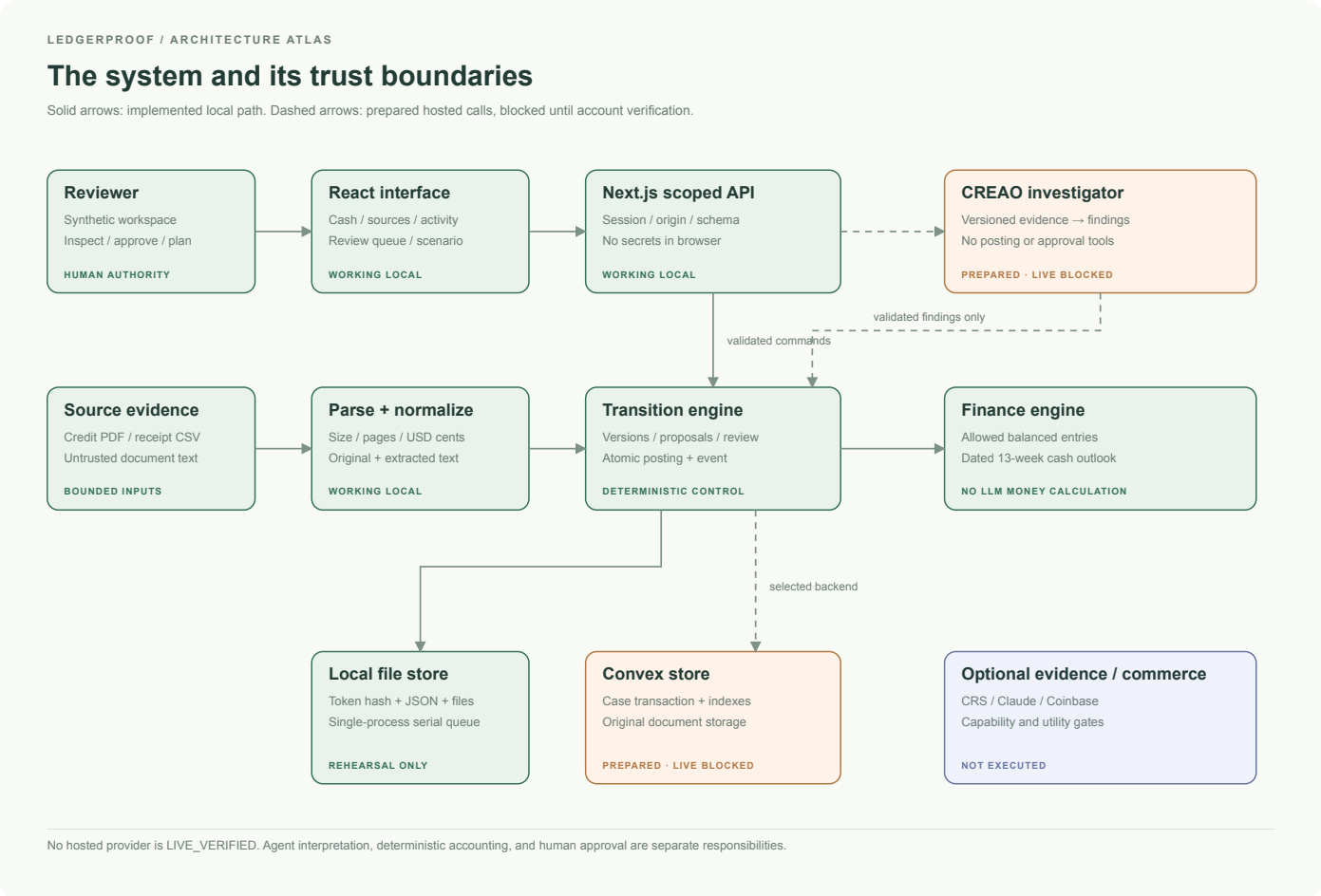
No sponsor was added merely to claim another track. No paid service was purchased, real money moved, customer contacted, public deployment made, or Devpost submission sent. Missing account access is summarized in ACTION-ITEMS.md, so you do not need to reconstruct the conversation.

Final verification: 20 unit/backend/provider-contract tests, 3 browser tests, typecheck and production build pass. Five diagrams, offline navigation/search/mobile behavior, and the 26-page PDF were checked. A local-only configuration doctor, changed-input test bundles, and an offline GitHub Actions workflow are prepared. Remote CI has not run; no source push occurred.

Architecture atlas

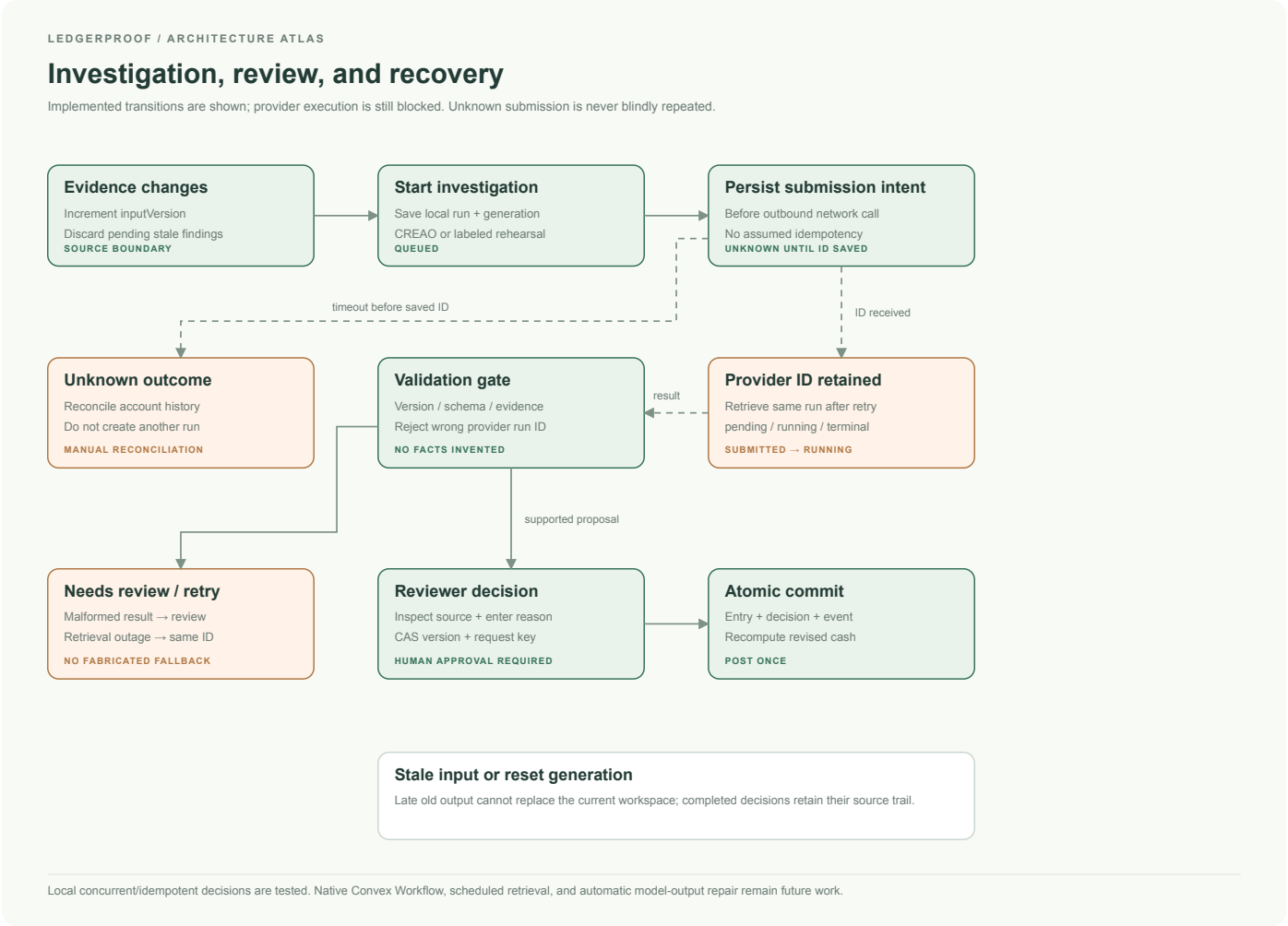
Each diagram has an editable Mermaid source, portable SVG and exported PNG under diagrams/. Regenerate SVG with `python3 scripts/render-diagrams.py`; regenerate briefing/PDF/PNG using the npm docs scripts. Green denotes a working local component, orange a prepared but blocked hosted path, and blue a future/conditional path. A box is not evidence of a live integration.

1. System context and trust boundaries



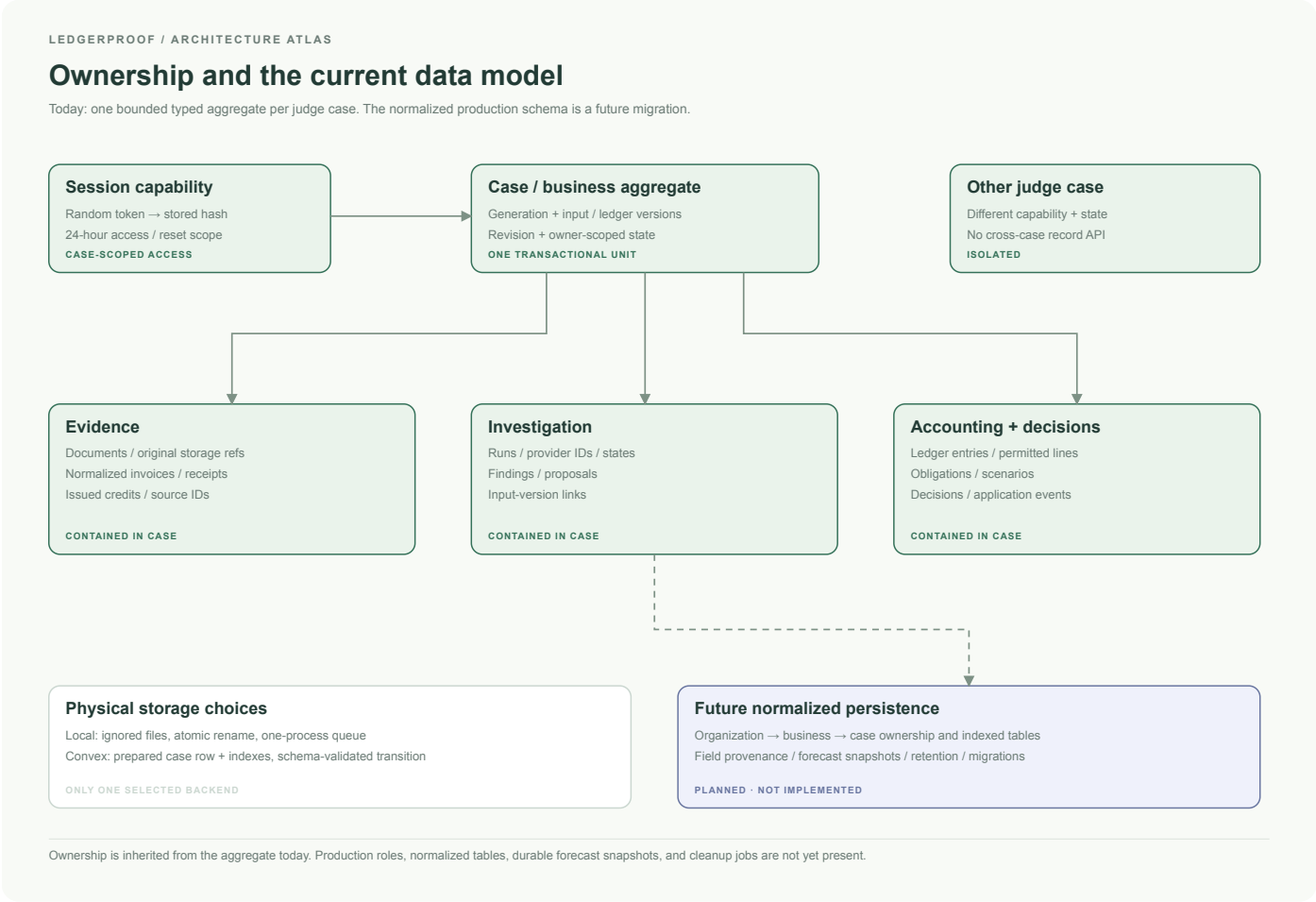
The browser handles presentation and review; the server handles scoped access and secrets. Documents are untrusted inputs. CREAO is intended to interpret evidence, while the deterministic transition and finance modules enforce posting and amounts. Local storage is tested; Convex/CREAO hosted paths are prepared but not verified. Editable source.

2. Investigation, review and recovery



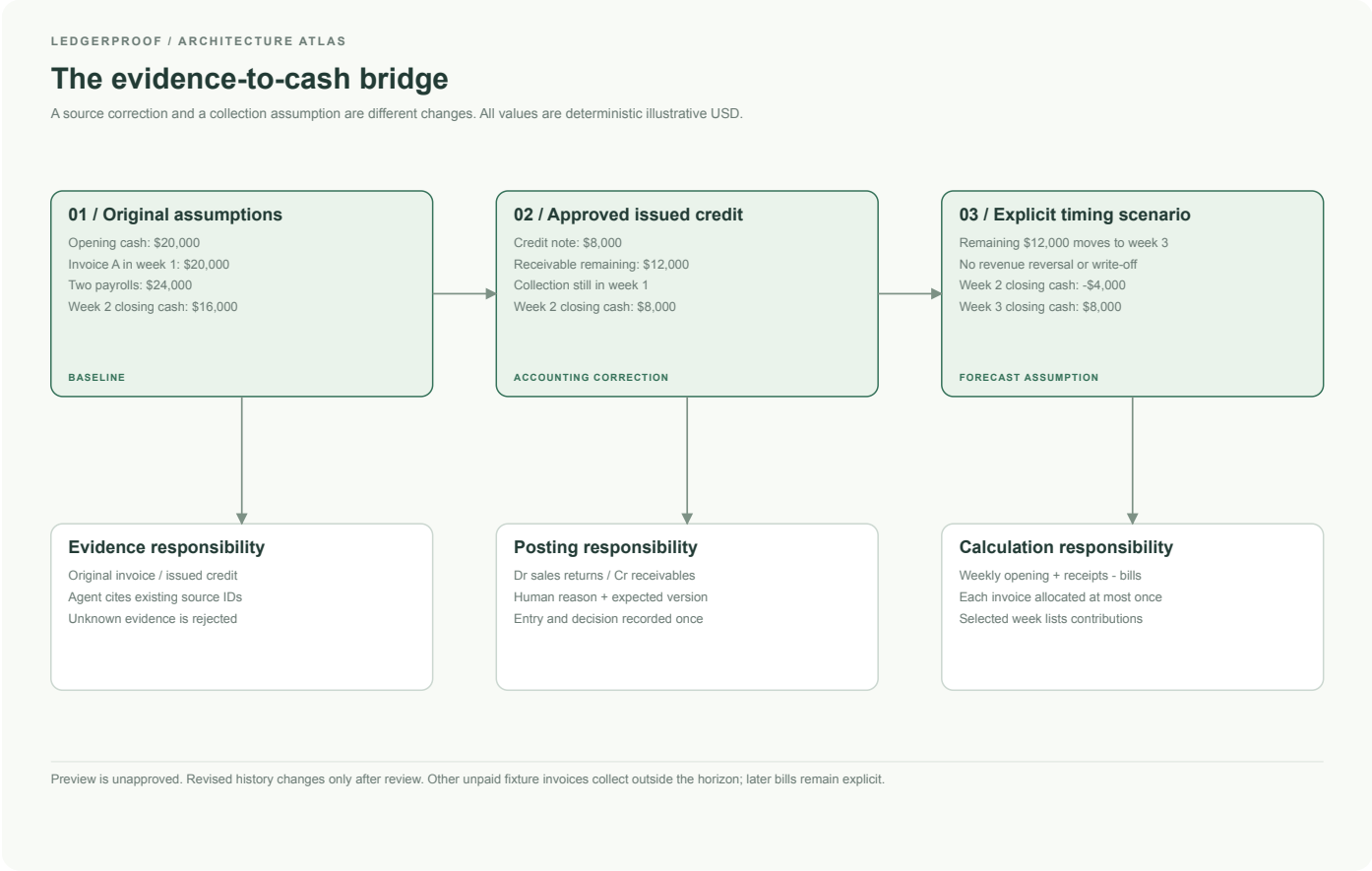
A saved provider ID is the recovery anchor. An ambiguous submission is not blindly repeated. Output must pass source/version checks before a supported proposal reaches the reviewer. Human approval is distinct from agent findings. Stale input and reset generations protect current state. Editable source.

3. Ownership and physical model



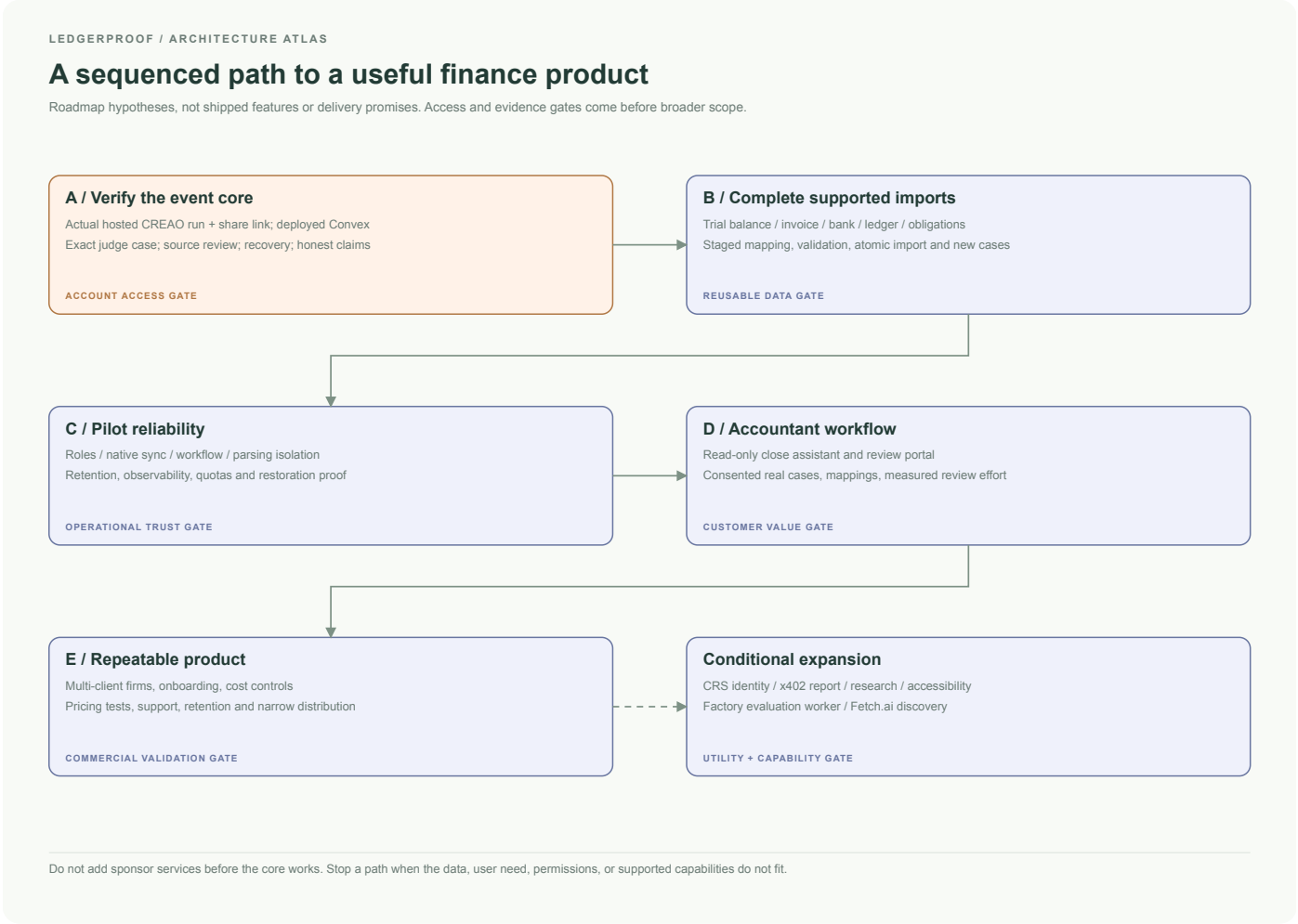
All current entities inherit one case's ownership in a bounded aggregate. Local file serialization and future Convex transactions are distinct implementations of the store interface. Production organization roles, normalized tables and persistent forecast snapshots remain future work. Editable source.

4. Evidence-to-cash calculation



The $8,000 credit is a reviewed accounting correction; moving collection to week 3 is a timing assumption. Neither change is a default probability or automatic write-off. The UI lets judges inspect contributions rather than trust a fixed reveal. Editable source.

5. Development gates



Verify the mandatory live core before expanding import breadth, reliability, accountant workflows, or conditional services. Phase estimates and acceptance criteria are in FUTURE-DEVELOPMENT.md. Editable source.

How LedgerProof works

The problem and intended user

A bookkeeper or fractional CFO serving a small creative agency wants to know whether payroll is covered. A spreadsheet can forecast a collection that is overstated, already partly paid, duplicated in the books, or subject to an unapplied issued credit. LedgerProof investigates those source facts and shows the consequence of a reviewed correction. It does not certify accounts, predict defaults, move funds, or automatically change an external accounting system.

The initial named community is independent bookkeepers and fractional CFOs serving small creative agencies. This is a product hypothesis, not validated customer research. No interviews or traction are claimed.

The complete user journey

1. **Start a workspace.** The server creates a random case ID and a high-entropy session token. The browser receives an HttpOnly, same-site cookie; the store retains a hash of the token. Workspaces expire after 24 hours and use synthetic records. Anyone given a workspace share link can review that case; this is a collaboration capability, not a production permission hierarchy.
2. **Inspect the original baseline.** The deterministic engine computes 13 explicit weeks from imported invoices, historical receipt allocations, opening cash, and dated obligations. Source dates and cents are authoritative; no language model calculates cash.
3. **Add evidence.** A bounded text-based credit-note PDF or one-row receipt CSV is parsed. The server checks the schema, invoice reference, amount, currency, document hash, size, and page limits. It stores the original and extracted text, increments the input version, and invalidates pending findings based on older evidence.
4. **Investigate.** The intended investigator is the hosted CREAO agent. The backend passes a complete versioned bundle of records and evidence. The agent identifies discrepancies, cites IDs, requests missing information, and returns findings. The app exposes no approval or posting tool to it. While credentials are unavailable, the user may explicitly choose local deterministic rehearsal. That branch is labeled and is not CREAO evidence.
5. **Validate the result.** Runtime schemas reject unexpected types, stale versions, unknown evidence, and record IDs outside the current case. Provider status/result IDs must match the stored provider run. Complete JSON output is required; prose is not silently interpreted as financial authority.
6. **Construct a correction proposal.** A supported credit-note finding can create a proposed debit to sales returns and credit to receivables. Receipt and duplicate findings use similarly narrow transaction templates. Source amounts are recalculated from normalized records; the agent cannot manufacture a numerical impact. Duplicate proposals reverse a source entry and preserve the original.
7. **Review the evidence.** The reviewer opens extracted text and the original PDF or CSV. Preview cash is labeled unapproved. The reviewer enters a reason and approves or rejects a specific proposal against the expected ledger version.
8. **Post atomically.** The state transition checks the version and idempotency key, validates balanced permitted accounts, updates the matching source status, writes the journal entry, records the decision, and appends an application event. In Convex this is intended to execute in a database transaction. In local rehearsal a single-process queue serializes file writes. Repeated or competing decisions cannot post the same correction twice in the tested local store.
9. **Change a scenario.** The reviewer may move an expected collection date. This changes cash timing without reversing revenue. An explicit reduced collection assumption is possible through the scoped API; it is a scenario, not a write-off or calibrated prediction.
10. **Explain and export.** The cash bridge lists a selected week's opening cash, receipts, obligations, and closing cash. The report export includes forecast, findings, run modes/references, and review events. JSON export is implemented; a customer-facing financial report PDF/XLSX is future work. The briefing PDF documents the product and is not a case accounting report.

The exact demonstration math

State	Week 1	Week 2	Week 3
Original baseline	$20,000 opening + $20,000 invoice − $12,000 payroll = $28,000	$28,000 − $12,000 payroll = $16,000	$16,000
Approved $8,000 credit	$20,000 + $12,000 receivable − $12,000 payroll = $20,000	$20,000 − $12,000 payroll = $8,000	$8,000

State	Week 1	Week 2	Week 3
Collection moves to week 3	$20,000 − $12,000 payroll = $8,000	$8,000 − $12,000 payroll = −$4,000	−$4,000 + $12,000 receipt = $8,000

Other unpaid example balances collect outside the forecast horizon. Later studio/software obligations are displayed in their real weeks. A deliberate duplicate ledger posting is represented separately from normalized invoices, so it does not cause the forecast to collect a second invoice. This is an intentional distinction between a messy source ledger and economic invoice identity.

Week boundaries are local date-only values, start-inclusive and end-exclusive. Calendar arithmetic avoids DST-hour errors. The shortfall indicator tests closing weekly cash below zero; it is not a guarantee that cash stays positive every day within the week. New matched receipts respect their transaction dates. New pre-horizon receipts explicitly reconcile opening cash; future receipts do not silently increase it.

Why three views of cash exist

Baseline preserves original imported collection assumptions and excludes newly approved credit corrections and scenarios. **Revised** reflects approved corrections and selected explicit scenario assumptions. **Preview** shows the effect of pending supported credits and remains separate from posted history. Each correction must pass review before it enters revised cash.

The model records historical invoice/receipt facts separately from collection assumptions. A credit changes the collectible invoice amount and the ledger after approval. A payment delay changes a forecast assumption only. A credit score or risk signal, if ever integrated legitimately, cannot automatically establish a write-off.

What each technical layer does

Layer	Goal	Actual implementation	Why the boundary matters
React interface	Make sources, review, and cash understandable	app/page.tsx, app/globals.css	Model findings do not bypass reviewer actions
Next.js server routes	Authenticate workspace, validate requests, keep secrets server-side	app/api/[...path]/route.ts	Browser never calls a sponsor with a secret
Runtime contracts	Validate uploaded, stored, and provider data	lib/contracts/index.ts	An apparently plausible JSON result is not automatically trusted
Document extraction	Convert supported sources to explicit facts	lib/documents.ts, unpdf, pdf-lib	Extraction coordinates are not invented; scans are rejected
Finance engine	Compute balances, allowed journal entries, forecast buckets	lib/finance/index.ts	Money stays integer cents, independent of an LLM
Transition engine	Enforce approvals, versioning, idempotency, reset behavior	lib/workflow.ts	One coherent mutation controls posting and decisions
Store	Persist scoped state and originals	lib/store.ts, convex/cases.ts	Local files are not mislabeled as Convex
Investigator	Interpret evidence and choose discrepancy findings	lib/providers/creao.ts	CREAO is substantive when verified; local rehearsal is separately labeled
Evidence/register docs	Tie claims to working execution and source requirements	docs/integration-status.md, SPONSOR-EVIDENCE.md	Configured code is not a successful integration

See SERVICE-REGISTER.md for every provider, including services not yet integrated.

State, ownership, and failure behavior

The current bounded workspace is one aggregate with nested typed entities. It contains business, invoices, receipt allocations, credits, sources, obligations, entries, scenarios, runs, findings, proposals, decisions, and events. Convex indexes locate the case and expiry. This makes approval atomicity easy to reason about for a small judge case. It is not a claim that the normalized schema in the full specification has been completed.

Evidence changes increment inputVersion. Approval increments ledgerVersion. Every mutation increments a workspace revision. Reset increments a generation. A result from an old generation is ignored; a result from old evidence is stale. A second reviewer must act against the same expected ledger version, preventing a decision based on obsolete balances.

Submission intent becomes unknown before calling CREAO. If the request times out before a provider run ID is safely saved, the application does not blindly submit another run. Account reconciliation is needed. If a run ID is saved, retrieval can resume against that same ID. Retrieval outages are retryable; malformed terminal results become needs_review. No webhook is enabled. Scheduled polling and bounded model-output repair remain unfinished; the present browser/API polling drives retrieval.

The agent instructions exclude document-borne commands. The code escapes source text and accepts only supported structured fields. Offline prompt-injection tests confirm that uploaded text cannot itself authorize posting; they do not establish that an untested hosted LLM will resist all injection attacks.

What has and has not been proved

Local tests cover exact forecast values, partial receipts, excess-credit exceptions, journal balance/account checks, competing decisions, session capabilities, reset/stale handling, actual PDF/CSV parsing, and provider-contract mocks. Browser tests exercise the full local review path, shared polling updates, unavailable CREAO, timeout rehearsal, and mobile layout. See EVALUATION.md for current counts.

Not proved: a live hosted CREAO run; a shared successful agent run; actual Claude use; deployed Convex transactions, subscriptions, or components; external sandbox data; Coinbase agent payments; real customer value; sponsor eligibility; production security or accounting completeness. There is no deployed URL, and no external accounting system has been altered.

Service-by-service implementation register

As of October 24, 2026. WORKING_LOCAL refers to observed local execution; it is not a sponsor integration status. UNTESTED means no successful provider execution, not implicit approval to claim a track. No paid account or subscription was created.

Runtime and infrastructure

Service/component	Goal	What I implemented	Evidence/status	Next gate
Next.js	Serve UI and secret-bearing API from one app	App Router page and scoped Node routes, same-origin mutations, private responses	WORKING_LOCAL; production build and browser tests	Choose hosting account/region, configure environment, authorize public deployment
React	Interactive judge flow	Four sections, source drawer, review reason, chart/cash bridge, scenario controls, sharing/reset	WORKING_LOCAL; desktop/mobile browser checks	Accessibility audit and real-user usability sessions
Convex	Central hosted persistence and atomic state transitions	Cases schema/indexes, guarded read/create/apply functions, file upload/read integration	BLOCKED; source typechecked, no deployment	Account project, URL and backend secret; deploy and verify transactions/storage
Convex live queries	Push case changes to collaborators	Not yet implemented; UI currently polls scoped HTTP state	NOT IMPLEMENTED; shared local polling is tested	Native subscription with real auth; two-session/reconnect proof
Convex Workflow	Durable scheduled execution and retries	Not installed/configured	NOT IMPLEMENTED	Add component only after runtime contract/account verification; never retry ambiguous non-idempotent submission
Local filesystem	Offline persistence while accounts are unavailable	Ignored .data JSON/originals, token hashes, single-process queue, atomic rename	WORKING_LOCAL; competing decisions and wrong-token tests	Keep a rehearsal tool; do not scale to multiple workers
Identity/session layer	Scope judge access without signup	Random bearer capability, HttpOnly cookie, share-link exchange, 24-hour access expiration	WORKING_LOCAL; isolation/share/reset tests	Production membership, role policy, revocation and retention cleanup

Core investigation and evidence

Service/component	Goal	What I implemented	Evidence/status	Next gate
CREAO	Core evidence investigation and evidence-bound challenges	Hosted-agent instructions, complete bundle, explicit developer-v2 or agent-v1 adapter, status retrieval, result validation	BLOCKED live; mock contract tests only	Create/save hosted agent, confirm account/API family and form fields, run changed evidence, retain ID/share link
Local investigation rules	Exercise review/forecast offline	Deterministic duplicate, partial, unmatched-receipt and credit rules	SIMULATED relative to hosted agent; local counts measured	Never use as CREAO submission proof
Anthropic/Claude	Meaningful extraction or reasoning with traceable model	No direct SDK call; possible account-supported selection inside CREAO	BLOCKED; no model trace	Verify model selection/output and evaluate actual benefit; only add direct extraction if necessary
Zod	Reject malformed data and unsupported output	Contracts shared by provider, state machine, parsers and store	WORKING_LOCAL; invalid enum/citation/version tests	Version migrations and explicit return schemas for expanded APIs
unpdf	Extract text from supported source PDFs	Signature/page/size checks, text extraction, resource cleanup	WORKING_LOCAL; generated PDF parser test	Process-isolated parsing, time/memory budgets, OCR gate for scans
pdf-lib	Produce internally consistent source fixtures	Invoice/credit PDFs generated from fixture definition; editable credit control	WORKING_LOCAL	Separate customer case-report PDF with proper metadata and disclosure
Finance engine	Reliable cash and posting calculations	Integer-cent outstanding balance, narrow entries, dated 13-week forecast, baseline/revised/preview	WORKING_LOCAL; exact cash/edge-case tests	General trial-balance/import support and accountant validation

CREAO owns interpretation; the finance engine owns amounts; the reviewer owns approval. Convex owns persistence when configured. The UI owns presentation. Maintaining those boundaries prevents an agent's plausible explanation from becoming an unauthorized accounting fact.

Conditional and adjacent providers

Provider	Potential job	What actually exists	Status	Fit/activation condition
CRS	Supported identity discrepancy investigation or a transparent collection scenario	No adapter/request/response	UNTESTED	Inspect sandbox catalog, provided identity type, and actual fields. No fictional business credit inference from consumer data
Coinbase for Agents + x402	Buyer agent purchases a bounded completed report from seller	Explicit unavailable commerce route; challenge reviewed	UNTESTED	Verify supported test tooling, buyer budget, payment validation, one receipt, recoverable delivery; separate from agency forecast
Bright Data	Essential current public counterparty evidence	No scraping code	NOT_TARGETED	Core must truly depend on permitted live public data; synthetic-ledger-only story does not satisfy that requirement
Factory Droid SDK	Independent reconciliation regression/evidence-pack worker	No SDK call	NOT_TARGETED currently; new conditional path	Meaningful runtime trigger/output with bounded access, after core works; developer use alone is insufficient
Fetch.ai	Discoverable bounded analysis agent for other agents	No agent registration or SDK	NOT_TARGETED currently; new conditional path	Genuine agent-to-agent request/result flow plus verified challenge rules
Figma	Real design iteration and traceable workflow	No Figma files, Make prototype, or Slides	NOT_TARGETED currently; full criteria retrieved	Actual intentional Figma use, accessible artifacts, required slides/stage log/embedded prototype, Edu verification
Microsoft	Production-minded agent evaluation track	Architecture, controls and roadmap documents	Eligibility UNTESTED; not an API integration	Actual end-to-end agent, measurable usefulness and customer evidence; Azure not inferred as required
Hyphenate	Controller-seat judging lens	Reviewable receivables workflow, honest failure modes and demo script	Eligibility UNTESTED	Real CREAO-powered flow and defensible evidence, plus event rules
SpacetimeDB	Alternative central shared state	No implementation	NOT_TARGETED	Only reconsider as an intentional Convex replacement, never a duplicate sponsor database

Additional guide sponsors are assessed in EVENT-STRATEGY.md. None has been added as a superficial dependency.

Engineering services and developer tools

GitHub repository: cloned from ArihantSwain/LedgerProof. Local source is the reviewable deliverable; no external repository publication is assumed. Git records and development provenance should remain honest about implementation dates and tool use.

Node/npm: dependency installation and lockfile; build toolchain. Playwright: actual headless Chromium checks and documentation rendering. Node test runner/tsx: finance, store, parser, and provider-contract tests. Python standard library: reproducible vector diagram generation. No Figma use is implied by hand-built SVG diagrams, and no sponsor SDK use is implied by generic tooling.

Google Drive connector: read the Coinbase challenge and Figma brief linked in the supplied guide. This was **research during development**, not a runtime LedgerProof data integration. No Drive files were edited, shared, or sent. Browser/search reads were used for public technical/event verification. They are not evidence of Bright Data, Apify, or a connected runtime web-research feature.

Secrets, cost, and failure ownership

CREAO_API_KEY, selected API family/base, agent ID, and actual input field belong on the server. Convex URL and backend secret are required together; the same secret must be configured in the backend environment. No key is present in .env.local because that file does not exist yet. A configured key does not change integration status to LIVE_VERIFIED automatically.

Every hosted provider call may consume credits or create an external effect. No paid subscriptions or real-money transfers have been initiated. Future cost controls should bound runs per case, bundle size, retries, concurrency, extraction pages, storage retention, and report generation. Cost targets must be measured from real provider receipts; there is no verified per-case cost today.

Future development: from judge demo to a receivables control product

Product direction

The strongest initial product is an evidence-to-cash workbench for independent bookkeepers and fractional CFOs who serve small creative agencies. It should earn a narrow seat: check whether receivables supporting a forecast are correct, prepare defensible adjustments, and let an accountable human decide. “Autonomous CFO” would overstate the scope and permissions of the present system.

The defensible asset is a traceable connection between source documents, normalized facts, corrections, reviewer decisions, and forecast effects. A generic finance chat interface is easy to replace. A reliable review workflow integrated into month-end close and weekly cash planning may be useful; that value still needs customer validation.

No timeline below is a delivery promise. Effort is an initial engineering estimate assuming one experienced builder and available accounts. Existing blocked integrations must be proved before expanding them. Do not add integrations to increase a sponsor count.

Phase gates

Phase	Intended outcome	Work	Exit evidence	Approximate effort
A: credible event core	Real investigator, honest live demo	Save CREAO agent, confirm API family/form, changed-input round trip, deployed Convex, exact judge path	Agent ID/run share link, two different outputs, source citations, hosted test receipts	3–8 focused hours after access; provider latency/account issues may extend
B: complete narrow imports	New judge inputs work without rewriting fixture	Opening balances, invoices, batch bank CSV, ledger and obligations import with explicit schema/mapping/preview	Balanced opening trial balance, atomic import, no duplicated IDs, parser rejection tests, one new independent case	3–6 builder days
C: dependable pilot	Bookkeeper can repeat weekly workflow	Workspace auth/roles, native sync, durable polling, data retention, parser isolation, monitoring	Reconnect/restart/concurrency tests, role denial tests, restore drill, documented operational runbook	1–3 builder weeks
D: useful accounting workflow	Reduce real review effort	Read-only accounting connector, review packets, historical statements, close checklist	Accountant-reviewed mappings, representative customer cases with consent, measured review-time improvement	3–6 builder weeks
E: repeatable product	Safe multi-client operation	Organization tenancy, quotas/billing, connector lifecycle, permissions, support tooling	Retained pilot users, cost/error budgets, end-to-end tenant isolation and deletion proof	6–12+ builder weeks

Immediate engineering priorities

Prove CREAO before elaborating it. Create one saved investigator with a clearly named bundle input. Confirm whether the actual account uses agent-v1 or developer-v2, then choose that family explicitly. Retain real status envelopes, model metadata if available, input/output, and share links with secrets removed. Show changed credit amounts and an evidence-bound challenge. No automatic fallback between API families. If the account cannot invoke externally, record the exact blocker rather than replace CREAO with local chat.

Make execution durable. Introduce a Convex Workflow or equivalent persisted scheduler for retrieval, validation and review pauses. Store bounded input hashes, provider IDs, attempts, deadlines and next retry time. Separate retryable retrieval from unknown submission outcome. Keep external actions idempotent where documented; otherwise reconcile. A workflow framework does not make an external non-idempotent call safe to repeat. Use notifications only with explicit user authority.

Finish imports before broadening finance. Build a staging area with field mapping, validation errors, cross-record reconciliation, and an all-or-nothing commit. Preserve originals and parser versions. Require explicit opening cash cutoffs, allocated receipt identity, and dated obligations. Handle partial allocations, credit-note identity, invoice aliases, duplicates, and overpayments as modeled exceptions. Support a deliberately limited USD service-business schema first; reject unsupported inventory or currency semantics.

Normalize persistence when aggregate limits matter. Move business entities to scoped indexed Convex tables without losing atomic approval semantics. Keep decisions, journal lines, source links and event appends in one transaction. Snapshot the exact forecast inputs/results used for a review. Add migrations, pagination, file cleanup, expiry cleanup, and cost/size bounds. Do not market an application event log as

immutable cryptographic evidence.

Harden extraction. Run untrusted PDF parsing outside the request process with time/memory/page budgets. Add OCR only after evaluating scans, and retain field provenance and uncertainty. A structured Claude extraction step is justified only if it improves the supported document cases and is actually measured. Make model/provider versions observable. Conflicting fields must remain review items rather than guessed "best" values.

New product paths worth exploring

Path	Customer job and why it fits	Smallest useful experiment	Success gate / stop condition
Read-only close assistant	Bookkeeper finds discrepancies before month-end close	Import one period, reconcile invoice/receipt/credit links, produce review packet	Accountant confirms findings and mappings; stop if data access produces more cleanup than benefit
Accountant–owner review portal	Owner approves understandable corrections while bookkeeper retains control	Separate preparer/reviewer roles and source-bound comments	Users complete review with fewer clarification rounds; stop if access or audit expectations exceed controls
Payroll reserve planning	CFO needs more than a zero-cash threshold	Reserve selector, daily/weekly cash bridge, transparent delay/expense grid	Decisions can be traced to selected assumptions; no probability/calibration claims
Evidence completeness monitor	Missing source documents delay close	Deterministic list of unsupported/unverified balances and requested documents	Fewer unsupported decisions, no false assertion that a missing item proves fraud
Counterparty identity checks	Conflicting legal/payment identity requires investigation	Only verified supported sandbox fields, source age, discrepancies and escalation	CRS product/identity type fits; stop if consumer fields must be misrepresented as business facts
Client remittance portal	Receipts arrive without invoice allocation	User uploads remittance advice and proposes allocation for review	Matching improves on representative evidence; no payment initiation or automatic posting
Dispute and credit-note workflow	Agency tracks why an issued credit changes collections	Credit identity, effective date, source chain, reviewer decision, scenario impact	Source-backed adjustment stays distinct from collection hypothesis
Cash sensitivity laboratory	Owner wants to understand several independent delays	Deterministic grid across amounts/dates/obligations with exact contributions	Values match engine invariants and clearly state assumptions; no unearned risk scores
Financial evidence API	Other tools need bounded, cited findings	Scoped read-only bundle/result API with schema versions and report hashes	Consumers can verify scope/citations and handle stale results; no arbitrary SQL or provider-secret exposure
Report-as-a-service commerce	A buyer agent needs one completed evidence packet	Fixed-price test report, bounded agent budget, verified x402 flow, order/delivery reconciliation	Authentic tooling and test receipt, no duplicate fulfillment; stop if it distracts from core readiness
Public-source collections research	A real investigation needs current public counterparty information	Permitted source retrieval tied to a specific evidence question	Bright Data/Apify is essential and live; stop if removing retrieval leaves the product equally useful
Accessibility-first cash briefing	Reviewer cannot comfortably use dense tables	Keyboard-first review and plain-language/source-linked briefing; voice only after user research	Task completion improves in observed use; do not add voice solely for a track
Reconciliation regression worker	Teams need repeatable proof after mappings change	Trigger bounded evaluation suite; independent worker assembles test diffs/evidence pack	Factory SDK actually executes runtime work; no book-writing permission and no replacement of CREAO investigation
Discoverable analysis agent	Another agent can discover and request the service	Fetch.ai discovery/message path around a bounded case bundle and reply	Working agent-to-agent exchange has a user need, verified SDK/runtime, and no duplicate backend

These are opportunities, not implemented features. The top expansion bets are imports, a read-only close assistant, payroll reserve planning, and an accountant review portal. Commerce, voice, and public web research should follow demonstrated demand.

Customer validation plan

Start with five independent bookkeepers or fractional CFOs who serve small creative agencies. Recruit only through authorized outreach; none has been contacted. Ask them to describe their last receivables discrepancy and payroll planning workflow before showing the demo. Record current tools, frequency, review duration, source availability, error cost, and approval responsibilities. Obtain consent before recording or handling real records.

Then run a moderated task using synthetic data first: find the credit, inspect source, decide, explain the cash effect, and challenge a finding. Observe completion, misunderstanding, and time; do not lead the participant to the expected answer. A later pilot may use redacted records under an appropriate agreement. Keep time saved, acceptance rate, false positives and data-cleanup effort separate. A positive interview is not a paying customer or proof of retention.

Measurement and evaluation

Metric	Definition	Initial target hypothesis
Task completion	Users complete source → review → explained forecast unaided	At least 4 of first 5 moderated users; revise based on observed friction
Supported-finding precision	Accountant-supported findings / all actionable findings on labeled cases	Set only after collecting a representative labeled dataset
Missed discrepancy rate	Known supported discrepancies not detected / labeled discrepancies	Stratify by type/document quality; do not hide misses in a single percentage
Review effort	Median minutes plus clarification rounds per case	Compare same case/workflow against actual baseline, not guessed labor cost
Correction integrity	Duplicate postings, stale approvals, cross-tenant mutations	Zero tolerated in required tests and observed pilot; incident process if violated
Agent usefulness	Reviewer can explain evidence and next action from findings	Qualitative rubric plus accepted/rejected reasons, independent of prose fluency
Runtime reliability	Terminal result or actionable failure within declared deadline	Measure queue, provider, validation and review latency separately
Cost per completed case	Actual provider + storage + compute costs including retries	Establish from receipts before pricing; no current measured figure

Expand evaluation beyond the current authored fixture: clean invoices, partial allocations, real issued-credit variants, false duplicate candidates, stale imports, conflicting references, scans, timeout recovery, and injection attempts. Split examples into development/evaluation sets. Record exact provider/model/prompt/schema versions, counts and uncertainty. Human approval remains necessary even with a strong evaluation score.

Production architecture direction

Use organization → business → case ownership, with viewer/preparer/reviewer/admin roles and a formal authorization matrix. Replace shared bearer editing links with revocable scoped invitations for production. Keep demo and customer environments separate. Add environment-specific keys, minimum data retention, encrypted storage managed by the selected platform, deletion workflows, original-file access controls, audit export and incident response. Security claims require verification; no certification is implied.

Operate with a Node-capable Next.js host and a genuine Convex deployment. Add native subscriptions or an authenticated server relay, background retrieval workers, limits per organization/provider, observability with sanitized request IDs, model cost budgets, and a deployment rollback plan. Test expiry and restore behavior. Avoid logging raw financial/identity responses or share tokens. A database transaction can make local posting atomic; it cannot make an external payment and local delivery globally atomic.

Commercial angle and distribution hypotheses

Possible packaging: a per-case investigation tool for independent bookkeepers; a recurring multi-client workspace for fractional CFO firms; or an evidence API/report service. Pricing, willingness to pay, market size and channel performance are unvalidated. Do not invent an addressable-market number or an hourly savings figure.

For a per-case hypothesis, contribution margin equals price minus actual extraction, investigator, storage/compute, payment fees, support, and retry costs. For a subscription, model included cases and guard against one heavy user consuming the whole allowance. Measure completion, monthly reuse, churn, support burden and source-import effort before adding automated billing. Accounting firms may be a better initial channel than direct sales to every small agency, but interviews must test that hypothesis.

Explicit stop conditions

Stop adding sponsor integrations if the mandatory live investigator or deterministic review path is not ready. Stop a credit-data path if supported fields do not fit the identity/business story. Stop scraping if evidence is non-public or behind a login/paywall. Stop autonomous posting/payment scope until a separately authorized design, identity policy and controls exist. Stop claiming statements are complete when the trial balance, cutoff or opening balances cannot be reconciled.

Recommended next release

Release 0.2 should prove live CREAO + Convex, finish a narrow reusable import pipeline, produce a complete scoped case report, and add resilient polling. Release 0.3 should support a real read-only pilot with roles, explicit retention, and accountant-reviewed mappings. Reserve secondary statements and payment commerce for later gates. The current product deserves a reliable receivables seat before it asks for a broader finance seat.

Cal Hacks 13.0 — strategy and additional paths

Source baseline: user-supplied hacker guide, preserved at sources/cal-hacks-13-guide.txt. Retrieved October 24, 2026: Cal Hacks official site, CREAO event pack, Coinbase challenge PDF, Figma challenge, and Fetch.ai event page. The Drive documents were read through the connected Google Drive skill; they were not edited. Some guide "more details" links were lost in pasted plain text.

Schedule and provenance

The supplied guide gives Sunday submission at 9:00 AM, judging 10:30 AM–12:30 PM, and requires presence for judging. In the October 23–25 weekend, that Sunday is **October 25, 2026, Pacific time**. This calendar mapping is an inference from the dated event context and supplied guide, not a fetched Devpost deadline field. Confirm the actual Devpost countdown upon return. The guide recommends an initial draft Saturday night. A draft was prepared locally; nothing was submitted.

The official site independently confirms in-person participation, a maximum of four teammates, and implementation during the event. Preserve real development chronology; do not change commit dates to manufacture eligibility. CREAO's page labels its pack October 23–24, while the guide includes Sunday judging; treat the guide/Devpost schedule as the event timing reference to check, rather than silently resolving that discrepancy. Fetch.ai's current page also differs from the guide in prize amounts; no expected winnings are calculated here.

Priority decision

1. **Core:** prove the CREAO investigator, exact cash math, source drilldown, review, recovery, and a runnable judge flow.
2. **Primary narrative:** a receivables controller assistant for Hyphenate; a production-minded finance agent for Microsoft; an AI-first receivables investigation for CREAO.
3. **Backend proof:** actually deploy and demonstrate Convex, with live state and durable execution rather than source-only claims.
4. **Conditional:** traceable Claude use; defensible CRS sandbox use; authentic test Coinbase Payment Agents flow.
5. **New paths:** evaluate Figma workflow and Factory runtime evaluation worker only when their real work exists. Do not rewrite the product to enter every track.

CREAO's event pack requires a hosted core agent, its ID/run share link, and visible execution. AI-First Disruptor is the clearest proposed fit. One-Person Company would need an actual coordinated organization of role agents, which is not implemented. Team Empowerment could fit a proven preparer/reviewer workflow, but track selection should follow the actual live product. Local rehearsal is engineering support, not the sponsor entry.

New sponsor paths reviewed from the guide

Track/provider	Product-fit path	Decision now	Evidence needed before any claim
Factory Droid SDK	Trigger a bounded reconciliation regression worker that produces diffs and evidence after an import/mapping change	CONDITIONAL NEW PATH	Actual SDK execution in running product, trigger, restricted workspace, plan/result artifacts; coding assistance alone does not qualify
Fetch.ai	Discoverable report/investigation agent with structured agent-to-agent requests	CONDITIONAL NEW PATH	Verified SDK/catalog rules and live requester/service exchange; no substitution for CREAO core
Figma	Traceable iteration of source review and cash-bridge interaction	CONDITIONAL NEW PATH	Real Design/Make/agent use, public artifacts, required slides, stage log, embedded prototype, ≤3-minute video and Edu verification
Capital One Nessie	Dedicated synthetic transaction-import adapter for sandbox accounting demonstrations	FUTURE ONLY	Actual supported sandbox API/data, useful mapping, reconciliation tests; do not call it real bank connectivity
Cartesia	Accessible spoken cash review for a user who needs voice as the main interaction	DEFER	Actual voice-centered customer task, working audio controls/latency/privacy; narration tacked on to dashboard is insufficient
Apify	Essential public counterparty source retrieval for an investigation	DEFER / SAME FIT GATE AS BRIGHT DATA	Live meaningful retrieval, provenance/failure handling, permitted sources; no duplicate scraper merely for another logo

Track/provider	Product-fit path	Decision now	Evidence needed before any claim
Elastic	Search a large multi-period evidence corpus by source/field	FUTURE ONLY	Real scale/search need, indexed scoped evidence and measurable retrieval quality; current 10-invoice case does not need it
Moss	Retrieval over prior reviewer decisions with scoped memory	FUTURE ONLY	Verified retrieval benefit, tenant filtering, stale evidence rules; current context is already a bounded bundle
Atlassian	Route reviewed finance exceptions into an existing operational workflow	UNASSESSED RULES	Full linked challenge not preserved in paste; then verify actual agent/workforce role and integration
Mana.space	Real planning/prototype workflow with shareable process artifacts	NOT USED	Actual usage that changed the build, public links and demo moment; cannot retroactively claim it
Notability	Authentic contemporaneous design/research notebook	NOT USED	Real evolving notes and feature use; generated docs alone are not Notability work
Meta	Collaborative owner–bookkeeper problem solving that materially strengthens connection	WEAK CURRENT FIT	User evidence and central AI collaboration, not generic finance automation with two accounts
Armadin security	Evidence prompt-injection and approval-boundary protections	RULES UNKNOWN	Published challenge plus independent attack/defense evidence; tests alone do not establish competition eligibility
Bitdeer Model Studio	Material extraction model serving role	DEFER	Actual model inference and evaluation; avoid a second inference service without demonstrated need
Visa	Bounded agent purchase/review workflow could touch commerce	WEAK CURRENT FIT	Genuine shopping/commerce problem and full rules; selling one report does not establish eligibility
SpaceXAI	Space-domain finance/research adaptation	NOT_TARGETED	Space-data and specified tool requirements do not fit this product; do not claim unperformed Cursor/API usage
Spacetime	Alternate central realtime architecture	NOT_TARGETED	Would replace Convex intentionally; no competing database added
Hardware / planet-first	No essential hardware/environmental role in receivables MVP	NOT_TARGETED	Do not fabricate relevance

These are evaluated paths, not integrations or submissions. New optional dependencies were not installed because their account contracts, utility and eligibility remain unproved.

Figma: changed from TBD to known requirements

The linked brief asks for a working demo with an honest workflow across ideation, prototyping, handoff, and code generation. Its specific deliverables include a short demo video, stage log (up to 200 words per stage), Figma Slides, accessible Figma artifacts/embedded prototype and repository, and Edu verification. The stage log should identify actual product use, elapsed minutes, a 1–5 assessment and downstream use. See WORKFLOW-STAGE-LOG.md for honest current history: **no Figma use has occurred**, so eligibility is not claimed.

An honest future Figma contribution could test a clearer source-review interaction with an accountant and carry that change into code. A decorative prototype created only to claim the track would not tell a convincing workflow story. Creating a Figma account, Edu verification, sharing artifacts, and making a video require account/human actions not available in this run.

Coinbase: primary brief reconciled with guide

The linked PDF supports the planned Payment Agents track: a bounded buyer can purchase the completed analysis through Coinbase agent tooling and x402. Its submission asks for a summary, reusable working example, architecture/controls/limits, and firsthand builder feedback. It permits clearly labeled simulation and does not score money spent. Actual account eligibility and the one-prize-per-person condition must be checked personally; the guide's aggregate prize wording does not override those terms. No eligibility, SDK execution, protocol interaction or settlement has been established here.

Keep buyer budget, quote, payment status, report delivery, and receipt reconciliation in a separate commerce state machine. Do not add simulated report revenue to the agency's payroll fixture. Avoid trading as a second primary direction. No real-money action is necessary or authorized.

Evidence capture checklist

Claim	Minimum artifact
CREAO does core work	Actual saved agent ID, successful run share link, changed-input result, live run in demo
Finance is correct	Independent expected manifest, exact changed-input tests, selected-week contribution bridge
Human control	Source inspection, reasoned approval/rejection, repeat/competing decision proof
Recovery	Real provider ID retained through interrupted retrieval; clearly separate simulated timeout
Convex is central	Dashboard deployment/function evidence, two clients, reconnect/restart, transaction and storage checks
Claude contributes	Actual model trace and output affecting supported findings/extraction, evaluation and limits
CRS is meaningful	Actual sandbox request/response, provided identity type, supported fields, visible defensible impact
Customer usefulness	Consented interview/task observations and measured comparison; no invented testimonials
Source chronology	Real local Git history, stage log, implementation dates, original guide/source pointers

Two pitches to prepare

For the finance judge: “We fill the receivables controller assistant seat. This agency expects $16,000 after payroll. An issued credit and collection timing change reveal a $4,000 gap. The investigator cites evidence; the human approves; the deterministic engine explains the cash.” Open with the shortfall moment and then show how it was discovered.

For the technical judge: explain the agent/accounting/approval boundary, case ownership, stale-generation guard, idempotent atomic posting, genuine provider status and recovery. Name which services actually ran. If CREAO is still blocked, disclose that before showing the rehearsal; no phrasing should imply a live integration.

Core actions for your return

Everything possible without account decisions was handled locally. No response is required while you are away. Start with the two account tasks below; optional sponsor tasks should not delay the core.

P0 — make the mandatory agent real

Where: agent.creao.ai, using your event-upgraded account. **Estimated human interaction:** 10–20 minutes plus provider execution time; not a guarantee.

1. Finish the fresh build of **LedgerProof Receivables Investigator**. Use CREAO-AGENT.md. Its job is evidence-backed reconciliation; it must not send email, approve corrections, post entries or transfer funds.
2. Save it as a reusable hosted agent, not just an unsaved conversation. If using the agent-v1 form, add a text input named **bundle** that accepts the serialized JSON evidence bundle. Retain the exact field name.
3. Locate external invocation settings. Confirm **which API family this account actually supports**. The app now supports explicit `agent-v1` or `developer-v2`; it never switches families automatically. If agent-v1 is supported, use base `https://agent.creao.ai/api/v1`; if developer-v2, use `https://developer.creao.ai/v2`. An agent ID/key from one family is not presumed compatible with the other.
4. Put the key and agent ID in LedgerProof/.env.local through your secure local/hosting environment. Never paste the key into chat or commit it. Set CREAO_API_FAMILY, CREAO_API_BASE and, for agent-v1, CREAO_INPUT_FIELD to match the confirmed account. Set CREAO_CONTRACT_VERIFIED=true only after checking that contract.
5. If supported, select a traceable Claude model and retain the actual model ID. Do not claim Anthropic otherwise.
6. Run the opt-in smoke script only with authorized event credits/account access: `node --env-file=.env.local --import tsx scripts/smoke-creao.ts`. Confirm changed $8,000/$6,000 credit inputs produce source-bound results, and run the UI flow with CREAO selected.
7. Capture the actual **agent ID and successful run share link**. Add them to README, SPONSOR-EVIDENCE.md and SUBMISSION.md. A run ID alone is not a shared run link. If the form/output differs, keep the real sanitized response and adjust the adapter explicitly.

Manual test inputs are ready in fixtures/creao/input-credit-8000.json and input-credit-6000.json. The exact machine-readable result contract is fixtures/creao/output-schema.json. These are fixtures and expectations, not live outputs. `npm run doctor` checks the current shell configuration; use Node's `--env-file=.env.local` form to load a local file.

Done when: hosted CREAO actually investigates changed inputs, valid cited findings enter the review queue, and the shared successful run is accessible to judges. Current status remains BLOCKED until then.

P0 — activate the real persistent backend

Where: your Convex account/project.

1. Run `npx convex dev` from LedgerProof and select/create the intended project using available event credits. Stop before any paid subscription or unintended project change.
2. Configure server-only CONVEX_URL in .env.local and the same strong random LEDGERPROOF_BACKEND_SECRET in .env.local and the Convex deployment environment. Keep secrets out of source/chat.
3. Deploy the prepared schema/functions to the authorized development deployment. Test create/read, original-file storage, changed evidence, concurrent decisions, reset and retrieval resume.
4. Verify shared updates in two browser contexts against the actual backend. Current UI uses polling; native subscriptions and a Workflow component are still not implemented and must not be claimed.

Done when: test receipts prove the hosted backend is central, durable and scoped. Local filesystem tests alone do not prove Convex behavior.

P0 — event participation and submission

Check the actual Devpost countdown. The supplied guide says Sunday 9:00 AM submission and 10:30 AM–12:30 PM in-person judging; the current weekend maps Sunday to **October 25, 2026, Pacific time**. Verify that mapping/countdown with the event site/app. Confirm participant eligibility, registration, teammates (maximum four), and actual implementation chronology. Be present for the full judging window as required by

the guide.

The draft is docs/SUBMISSION.md; the two-minute story is docs/DEMO.md; source and evidence are indexed in START-HERE.md. A Saturday draft is recommended by the guide. **No Devpost submission has been made.** Source publication, a public deployment, video recording/public sharing, and submission still need your concrete account decisions/authorization. "Any command" was used for local work; it was not interpreted as permission to spend money, contact people, trade, or submit on your behalf.

P1 — only if the core is already working

Action	What only you can supply	What would make it useful
Confirm real user problem	Your lived experience or a consenting independent bookkeeper/fractional CFO serving creative agencies	Observed workflow/task feedback; no fabricated interview
CRS capability gate	Event sandbox credentials, provided test identity, catalog access	Supported fields and identity type genuinely inform a visible investigation/scenario
Coinbase test tooling	Eligible account confirmation, authorized test keys/tooling, environment choice	Actual bounded buyer-agent/x402 report purchase with recoverable delivery; no real funds needed
Figma path	Edu verification and access to Design/Make/Slides	Real iteration reflected in code, accessible artifacts, honest stage log and demo video
Factory/Fetch.ai	Verified event SDK access only if pursuing a real runtime role	Independent evaluation worker or genuine agent-to-agent exchange; never replace CREAO core
Hosting	Preferred existing account/domain and publishing decision	Deploy a tested build with real backend and exact APP_ORIGIN; record real URL

What you do not need to supply now

No real bank credentials, customer financial records, personal credit identity, payment authority, broad email access, or paid subscription is needed for the synthetic judge core. Bright Data, Spacetime, voice, hardware and other extra tracks are not blockers.

Copyable handoff after account work

Tell the next coding run: "CREAO is saved. The nonsecret API family is ___, agent ID is ___, input field is ___, and the run share link is ___. Secrets are configured locally, not in chat. Convex project/URL is configured. Verify actual execution, update integration-status, finish outstanding core gaps, then rerun the full judge flow." Do not say the integration works until the execution artifacts support it.

Evaluation

Executed October 24, 2026. None of these checks establishes live provider readiness.

Check	Actual result
`npm test`	20/20 passing finance, workflow, PDF/CSV parser, capability, concurrent local-store and mock-provider contract tests
`npm run typecheck`	Passed
`npm run build`	Passed with Next.js 16.4.3, static page and dynamic scoped API
`npm run fixtures`	Generated 10 invoice PDFs, credit PDF, bank/ledger/obligations CSVs
`npm run test:e2e`	3/3 passing in 6.8 s on final autonomous pass against production build: credit upload → rehearsal → review → forecast; shared browser updates and unrelated isolation; blocked CREAO + simulated timeout resume; mobile fit/navigation
`node --import tsx scripts/evaluate.ts`	3/3 authored anomalies detected, 0 missed, 0 false-positive anomalies; 1 benign partial payment finding; validation failures 0; local elapsed 1.541 ms
npm dependency audit at installation	0 reported vulnerabilities
`npm run fixtures:creao`	Generated two-document $8,000/$6,000 credit evidence bundles and exact JSON result schema, with no provider calls
`npm run doctor`	Reported missing CREAO/Convex/account environment actions without printing secrets or making network calls
`npm run docs:build` / `docs:export`	Generated five SVG/PNG diagrams, a 10-section offline HTML briefing and a 26-page PDF; navigation/search/empty-state/mobile checks passed

Browser screenshots were inspected for desktop dashboard and mobile review. Browser artifacts are ignored under test-results/. Initial browser attempt exposed a test synchronization issue (navigation before upload completion); corrected by waiting for the upload response and testing the production build. Disk exhaustion during install left a damaged compiler, which was replaced before successful checks.

Declared synthetic evaluation set: 10 invoices with normal fully matched receipts, one partial payment, one unmatched receipt, one duplicated ledger entry, plus an optional uploaded issued credit. Expected results live independently in fixtures/expected.json. This small authored set is not production performance evidence.

Required checks: exact cash transitions; date boundaries/rollover; balance/account enforcement; overpayment exceptions; preview separation; amount changes; timing preserves revenue; no risk write-off; duplicate and competing decisions; stale/reset behavior; scoped sessions; citation/enum rejection; prompt injection remains data; explicit blocked provider status; source-upload browser path. Browser test source is in tests/browser.

Measured offline rule sample: one authored synthetic agency, 10 invoices, 7 clean negative invoices after credit introduction, three seeded anomalies (credit, unmatched receipt, duplicate posting), and one legitimate partial payment. See evaluation-results.json. This is a small deterministic rule check, not a general accuracy benchmark or evidence of CREAO quality. The explicit 1.541 ms measurement excludes network/model/extraction time.

Live CREAO extraction/matching accuracy, missed errors, false positives, model latency, share links, Claude trace, Convex concurrency/subscriptions, and sponsor sandbox evidence: **not measured**. Customer interviews: none. No fabricated customer feedback or benchmark percentage.

Autonomous architecture review added regression tests for dated newly posted receipts and fractional-dollar display, plus mock HTTP tests for explicit CREAO API-family separation, mismatched base rejection, structured text parsing, wrong-run rejection, stale result rejection and unavailable credentials. These are offline contract checks, not successful provider calls. Malformed terminal results now escalate to review rather than repeating retrieval indefinitely. A GitHub Actions workflow was prepared but has not executed remotely because source has not been pushed.

Workflow stage log — honest draft

Prepared for traceability; **not a Figma-track eligibility claim**. Figma's linked brief asks for short stage records including product, purpose, minutes, rating and downstream use. No Figma product has been used. Exact stage durations were not instrumented, so they are not invented. This file can be extended with actual later work, without rewriting earlier history.

Ideation

User supplied the LedgerProof build specification: verify receivables behind a 13-week payroll cash forecast. It defined mandatory CREAO, preferred Convex, review controls, provider capability gates, and exact synthetic cash values. The coding agent translated that brief into a local implementation and explicit integration register. Tools: user-authored specification, Codex, local Markdown. Figma product: none. Minutes: not instrumented. Figma rating: not applicable. Output: preserved specification, accounting assumptions, contracts, fixture/expected manifest and phased scope. Customer problem remains a hypothesis; no interview is claimed.

Prototyping

The prototype was built directly in React/CSS with a source drawer, cash chart/bridge, activity and approval queue. Generated PDFs/CSVs supported a real local upload path. Playwright exercised desktop/mobile layouts and sharing/isolation. Source text stays escaped and human approval remains explicit. Tools: React, Next.js, pdf-lib/unpdf, Playwright and Codex. Figma product: none. Minutes: not instrumented. Figma rating: not applicable. Output: working local rehearsal plus observed screenshots; no separate Figma prototype exists.

Handoff

Shared runtime contracts connect parsed sources, provider output, state transitions and finance. CREAO-AGENT.md defines the hosted investigator's role, input and exact output. Service/status registers distinguish prepared source, local tests and absent hosted execution. The new architecture atlas, service explanation, roadmap and action list make gaps reviewable without the original conversation. Tools: TypeScript/Zod, Markdown, generated SVG/HTML/PDF. Figma product: none. Minutes: not instrumented. Rating: not applicable. Output: reproducible code, docs and explicit account actions; no design artifact is falsely attributed to Figma.

Code generation and verification

Codex implemented source locally after cloning the empty repository. Tests verify finance/state invariants, actual limited PDF/CSV extraction, local concurrent decisions and provider-contract mocks. Browser checks cover the end-to-end rehearsal. Disk exhaustion damaged a dependency; clearing a disposable cache and replacing the compiler enabled a successful build. Browser test timing was corrected to await upload completion. Architecture review found and fixed future receipts incorrectly entering opening cash. Tools: Codex, npm, Node/tsx, TypeScript, Next.js and Playwright. Figma product: none. Minutes: not instrumented. Rating: not applicable. Output: tested local code; hosted integrations remain blocked.

Later genuine Figma work, if any

Append actual product/file links, minutes, 1–5 assessment, observed friction and the specific change carried into code. Complete the required Slides and public prototype/embed deliverables only if the track is pursued. Do not backfill claims of usage or measured time savings that did not occur.

Sources, decisions, and confidence

Retrieved/reviewed October 24, 2026. Facts from source documents are distinct from implementation evidence. No account capability is inferred from a documentation page. No event eligibility is inferred from a technical integration alone.

Source	What was actually available	How used / limitation
User LedgerProof specification	Complete pasted implementation brief, preserved as BUILD-SPECIFICATION.md	Scope, exact forecast values, mandatory CREAO, accounting/approval rules and provider gates
User Cal Hacks 13.0 guide	Complete pasted plain text, preserved at sources/cal-hacks-13-guide.txt	Sunday schedule, live/in-person judging, sponsor summaries; some embedded links/images lost in paste
Cal Hacks official site	Public page retrieved	In-person, team maximum and event-start implementation requirements; no authenticated Devpost countdown
CREAO CalHacks2026 pack	Search-indexed official page retrieved; direct open failed	Core hosted agent / actual ID and share link / visible run; track fit. Page dates differ from Sunday guide context
CREAO developer v2 create and get	Public indexed documentation	Prepared developer-v2 adapter contract; live account and exact successful output not verified
CREAO API Trigger technical page	Public indexed documentation for another event	Prepared optional agent-v1 contract only; **not a Cal Hacks eligibility authority**, no automatic endpoint fallback
Convex Node client, mutations, workflows	Official technical pages retrieved	Backend design and future durability/subscription direction; no deployment receipt
Coinbase CalHacks PDF	Connected Drive readable text, title Coinbase CalHacks Challenge 2026.pdf, modified October 21, 2026	Primary Payment Agents/submission/eligibility source. No Coinbase account or SDK execution inferred
Figma challenge	Connected Drive readable text, title Cal Hacks <> Figma, modified October 24, 2026	Replaced TBD with actual stage-log/slides/prototype/video/Edu requirements; no actual Figma use
Fetch.ai CalHacks page	Public official event page	New conditional agent-discovery path; prize/address details differ from supplied guide, so no payout forecast
CRS sandbox entry	Direct page open unsuccessful; guide provides description	Catalog/test identities remain unavailable; no field or identity-type assumptions
Factory docs	Official plans page discoverable; guessed SDK overview route unsuccessful	No SDK contract or execution claimed. Only guide-grounded conditional worker design
Repository code/tests	Local inspected source and executed checks	Source of actual implementation/status claims; see EVALUATION.md

Architecture decisions

One investigator boundary: preserve CREAO for substantive investigation; local rules are explicitly rehearsal. Do not use another provider to pretend the mandatory dependency works.

One selected hosted database: Convex remains central by design. Do not add Spacetime just to collect a track. The local filesystem is an offline rehearsal implementation of the store interface and is disabled by default for production hosting.

Money stays deterministic: source-backed amounts, integer cents, dated obligations, narrow balanced entries, explicit exceptions. The investigator supplies cited interpretations rather than authoritative cash numbers.

Human authority is explicit: interpretation, proposal creation, approval, posting and scenario change remain distinct operations. Approval persists with its entry and event. No external book writes or payments are exposed to the agent.

Provider families are explicit: support developer-v2 and agent-v1 as separate selected adapters behind one interface. Never mix casing, forms, IDs, bases or result assumptions. Account confirmation still gates live calls.

Failed verification changes the claim: a malformed terminal result becomes needs_review; a failed retrieval retries the same ID; an ambiguous submission requires reconciliation. Public documentation, credentials, and mock responses never set LIVE_VERIFIED.

New paths must earn their place: the guide reveals possible Factory, Fetch.ai and Figma paths. They remain conditional because no meaningful runtime/tool workflow has been executed. Optional research/voice/commerce must not displace the receivables core.

Unknowns deliberately preserved

Actual CREAO agent ID/share link/API family/input form; model trace; Convex project/deployment; current Devpost countdown and participant eligibility; full linked Hyphenate/Atlassian briefs; CRS catalog/test identities; sponsor stacking limits; customer feedback; hosting account/region; real provider costs/latencies. These are ACTION-ITEMS, not missing values to fabricate.